

Balancing Accuracy and Environmental Efficiency in Financial Sentiment Models

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ABSTRACT

This project investigates how deep learning architectures differ in energy consumption when performing financial sentiment classification, with the main goal of identifying a balance between predictive accuracy and environmental cost. SimpleRNN and LSTM models were trained and evaluated while tracking carbon emissions with CodeCarbon on an Apple M2 Pro system. Results show that energy use depended strongly on hardware architecture interaction rather than parameter count alone. In particular, LSTM models, although more complex, produced lower emissions than SimpleRNN models due to faster execution and fewer training epochs. Among the configurations tested, the LSTM models provided the strongest balance between predictive performance and carbon efficiency, with minimal differences observed between the 16 and 32 unit configurations. These findings show that choosing architectures with awareness of hardware and training behavior can reduce environmental impact while maintaining necessary predictive performance

Index Terms— financial sentiment analysis, recurrent neural networks, LSTM, carbon emissions,

1. INTRODUCTION

Financial sentiment analysis is the process of organizing financial text according to whether the wording is negative, neutral, or positive. Prior work by Malo et al. [1] demonstrated the importance of detecting sentiment in financial texts for economic analysis. In this project, financial sentiment classification is used as a test setting to investigate how different recurrent neural network (RNN) architectures consume energy on specific hardware, with the goal of identifying a practical balance between predictive performance and environmental cost. A Simple Recurrent Neural Network (SimpleRNN) and a Long Short Term Memory network (LSTM) were trained on the Financial PhraseBank dataset [2] and CodeCarbon [3] was used to estimate the carbon emissions from model training.

During the original comparison between the models, the LSTM model produced much lower emissions during training, motivating the central research question of this project. This was unexpected because a more complex model is often assumed to require more energy. Additional experiments showed that LSTM models produced lower emissions because they trained faster and required fewer epochs under early stopping. Increasing the number of LSTM units did not significantly change emissions or performance, suggesting that model size had limited impact within this architecture under the same training conditions. This shifted the research question from simply identifying the most accurate model to understanding how architecture interacts with hardware efficiency and training behavior, and whether a practical balance can be identified between predictive performance and carbon emissions.

This question is important because model selection in machine learning depends not only on prediction accuracy but also on hardware efficiency. In a real world setting, companies may also need to consider energy costs, hardware availability, and sustainability goals or restraints. Therefore, this project evaluates how RNN architectures differ in energy consumption on Apple Silicon hardware and identifies what achieves a meaningful balance between prediction quality and environmental efficiency.

2. METHODOLOGY

2.1 Dataset

For the financial sentiment analysis, the Financial PhraseBank data set was used, containing 2,264 labeled financial news sentences across three sentiment classes (negative, neutral, positive), with neutral representing the majority class. The dataset was split using stratified sampling into training (70%), validation (15%), and testing (15%) sets.

2.2 Text preprocessing

Text data was converted into integer token sequences using a Keras TextVectorization layer with a vocabulary size of 10,000 and a fixed sequence length of 40. Each model incorporated a 32 dimensional embedding layer trained jointly with the classifier. To ensure fair comparison, preprocessing remained constant across all experiments.

2.3 Model architectures

Two recurrent architectures were compared: SimpleRNN and LSTM. To identify a potential sweet spot between predictive accuracy and carbon output, two unit sizes were evaluated for each architecture. Four final model configurations were compared:

- SimpleRNN (16 units)
- SimpleRNN (32 units)
- LSTM (16 units)
- LSTM (32 units)

Each model used the same structure:

- TextVectorization layer
- Embedding layer (32 dimensions)
- Recurrent layer (SimpleRNN or LSTM)
- Dense softmax output layer (3 classes)

Models were trained using the Adam optimizer and sparse categorical cross entropy loss. Early stopping based on validation loss was applied to prevent unnecessary computation and ensure emissions estimates reflected efficient training behavior.

2.4 Carbon emissions measurement

CodeCarbon was used to estimate carbon emissions during model training. The tracker was started immediately before training and stopped after, recording total training time, average epoch time, number of epochs, and estimated emissions (kg CO₂) for each model. Because these emissions are estimates rather than direct measurements, results were interpreted in the context of hardware detection. CodeCarbon reported no GPU usage and based estimates on CPU and RAM activity while TensorFlow identified both CPU and GPU devices on the Apple M2 Pro system. Therefore, emissions estimates were interpreted as differences in runtime and hardware execution behavior rather than exact energy usage.

3. RESULTS

3.1 Initial model behavior

The SimpleRNN established a reference point for comparing runtime behavior, accuracy, and emissions across architectures on the Apple M2 Pro system. While SimpleRNN achieved slightly higher accuracy, the LSTM models consistently trained faster and produced substantially lower emissions despite having more parameters. This contrast suggested that factors beyond model size and complexity played a role in determining overall performance.

3.2 Timing analysis

To ensure emissions estimates reflected full training behavior, all models were rerun from a clean kernel under identical training conditions.

The epoch timing analysis represented in Figure 1 showed a major difference between the SimpleRNN and LSTM models. The SimpleRNN models required approximately 34–35 seconds per epoch, whereas the LSTM models completed epochs in under 2 seconds. These runtime differences indicate that backend execution efficiency, rather than parameter count alone, was the dominant factor influencing emissions.

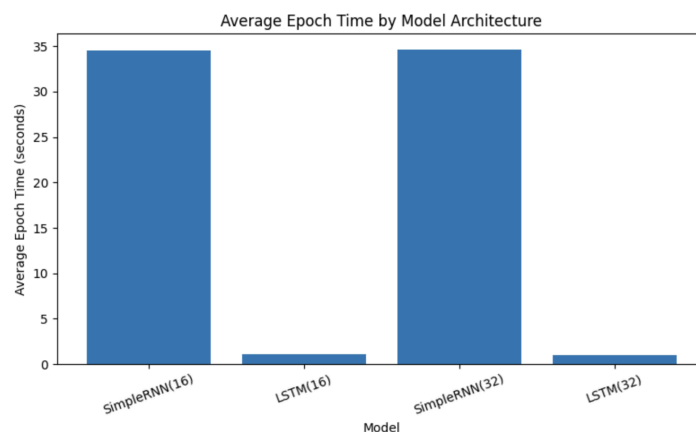


Figure 1.

3.3 Final model comparison

The final comparison of the four model configurations in Table 1 illustrate how architecture choice influenced both runtime driven emissions and predictive performance under early stopping conditions. Additionally, early stopping resulted in different numbers of training epochs across models, which directly influenced both runtime and emissions..

Model Ran	Test Accuracy	Training Time (seconds)	Epochs Completed	Average Epoch Time (seconds)	Emissions kg CO_2	Parameters
SimpleRNN(16)	0.75	172.55	5	34.5	0.001002	320,835
LSTM(16)	0.63	2.23	2	1.08	0.000013	323,187
SimpleRNN(32)	0.72	69.33	2	34.64	0.000402	322,179
LSTM(32)	0.62	2.06	2	0.99	0.000012	328,419

Table 1.

These results show that although the SimpleRNN(16) model achieved the highest test accuracy, it required the longest training time and produced the highest emissions. In contrast, the LSTM configurations showed similar patterns, producing the lowest emissions and fastest training time compared to SimpleRNN, but with lower predictive accuracy.

4. DISCUSSION

The most important finding is that environmental cost was not determined by parameter count alone. Although LSTM models had more parameters than SimpleRNN models, they produced substantially lower estimated emissions because they trained faster and required fewer epochs under early stopping. This suggests that model efficiency depended more on architecture hardware interaction than on model size alone.

TensorFlow detected both CPU and GPU devices, while CodeCarbon estimated emissions mainly from CPU and RAM usage, suggesting that TensorFlow likely used Metal acceleration during training. This

supports the idea that the most energy efficient model is not always the smallest model, but the one best suited for the hardware used.

The results also showed a practical accuracy and sustainability tradeoff. SimpleRNN(16) achieved the highest accuracy but produced the highest emissions, while LSTM models produced the lowest emissions but reduced predictive performance. The LSTM models provided a strong balance between prediction accuracy and environmental efficiency, with only minor differences between the 16 and 32 unit configurations. This suggests that increasing the number of units within the LSTM architecture had limited impact on both emissions and performance under the same training conditions.

This type of analysis is important in machine learning because model selection should consider computational efficiency and environmental impact alongside predictive performance. Although emissions values from CodeCarbon are estimates and may not fully capture GPU activity through the Metal backend, the consistent measurements across models supports the conclusion that informed hardware architecture selection can substantially reduce training emissions while maintaining necessary predictive performance.

5. CONCLUSION

This project showed that different RNN architectures can produce very different environmental costs on the same task depending on how they interact with the underlying hardware. Although the LSTM models had more parameters than the SimpleRNN models, they trained substantially faster on the Apple M2 Pro system and in turn produced much lower estimated emissions. This demonstrates that training emissions were primarily driven by hardware efficiency rather than architecture size alone.

The results identified a practical trade off between predictive performance and sustainability. SimpleRNN(16) achieved the highest accuracy but required the longest training time and produced the highest emissions. The LSTM models produced much lower emissions, with only minor differences between the 16 and 32 unit configurations, suggesting that model size had little effect within the LSTM architecture. Although the LSTM models did not achieve the highest accuracy, their performance remained comparable, making them a more efficient choice when considering both accuracy and environmental cost.

These findings show that model selection should consider hardware behavior, runtime efficiency, and environmental impact alongside predictive performance. In settings such as financial sentiment classification, slightly lower accuracy models may be preferred if they substantially reduce computational cost and emissions. This is increasingly important as organizations face growing expectations to manage the energy waste of machine learning systems.

Future work could extend this analysis by evaluating additional architectures such as GRU and by repeating the experiments across different hardware environments.

6. REFERENCES

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